

Solutions to Problem Set 3

1. (*) The Monty Hall problem

[4 Points]

In a game show, you have to choose one of three doors. Behind one door is a new car, and behind the other two doors are goats. After you make your choice, the host reveals one of the other doors, which always has a goat behind it. You are then given the option to switch your choice to the remaining unopened door.

- (a) Given that the host has opened a door revealing a goat, should you switch your choice to increase the chances of winning the car? Please explain your reasoning.
- (b) Let's call the two goats Bill and Nan. Given that the host has opened a door revealing a goat, and shows Bill with a probability of b , following the strategy in (a), what is the probability of winning the car?
- (c) If the host opens a door randomly, which strategy would you take to maximize your chances of winning the car?

- (a) WLOG, suppose that I choose door 1 (D_1) as my initial choice, the host opened door 2, D_2 , if I stick with my choice, the probability of winning a car: By Bayes theorem,

$$\begin{aligned}\mathbb{P}(D_1 = \text{car} | D_2 = \text{goat}) &= \frac{\mathbb{P}(D_2 = \text{goat} | D_1 = \text{car})\mathbb{P}(D_1 = \text{car})}{\mathbb{P}(D_2 = \text{goat})} \\ &= \frac{\mathbb{P}(D_2 = \text{goat} | D_1 = \text{car})\mathbb{P}(D_1 = \text{car})}{\mathbb{P}(D_2 = \text{goat} | D_1 = \text{car})\mathbb{P}(D_1 = \text{car}) + \mathbb{P}(D_2 = \text{goat} | D_1 = \text{goat})\mathbb{P}(D_1 = \text{goat})} \\ &= \frac{1 \times 1/3}{1 \times 1/3 + 1 \times 2/3} = 1/3.\end{aligned}$$

Hence, I would like to switch my choice, since $\mathbb{P}(D_3 = \text{car} | D_2 = \text{goat}) = 1 - 1/3 = 2/3$.

- (b) From (a), I will switch my choice, hence the probability of winning a car is

$$\begin{aligned}\mathbb{P}(D_3 = \text{car} | D_2 = \text{Bill}) &= \mathbb{P}(D_1 = \text{Nan} | D_2 = \text{Bill}) \\ &= \frac{\mathbb{P}(D_2 = \text{Bill} | D_1 = \text{Nan})\mathbb{P}(D_1 = \text{Nan})}{\mathbb{P}(D_2 = \text{Bill} | D_1 = \text{car})\mathbb{P}(D_1 = \text{car}) + \mathbb{P}(D_2 = \text{Bill} | D_1 = \text{goat})\mathbb{P}(D_1 = \text{goat})} \\ &= \frac{1 \times 1/3}{b \times 1/3 + 1/2 \times 2/3} = \frac{1}{b+1},\end{aligned}$$

since $\mathbb{P}(D_2 = \text{Bill} | D_1 = \text{goat})$ ensures that there is only a door with goat behind, and with probability $1/2$ it is Bill.

- (c) Of course, you wish the door opened by the host randomly revealed a goat, otherwise you lose immediately,

$$\begin{aligned}\mathbb{P}(D_3 = \text{car} | D_2 = \text{goat}) &= 1 - \mathbb{P}(D_1 = \text{car} | D_2 = \text{goat}) \\ &= 1 - \frac{\mathbb{P}(D_2 = \text{goat} | D_1 = \text{car})\mathbb{P}(D_1 = \text{car})}{\mathbb{P}(D_2 = \text{goat} | D_1 = \text{car})\mathbb{P}(D_1 = \text{car}) + \mathbb{P}(D_2 = \text{goat} | D_1 = \text{goat})\mathbb{P}(D_1 = \text{goat})} \\ &= 1 - \frac{1 \times 1/3}{1 \times 1/3 + 1/2 \times 2/3} = 1/2.\end{aligned}$$

No matter you decide to switch or not, the (conditional) probability of winning will be the same.

Remark: For question (a): Intuitively, since I randomly choose a door among three, and stick with my choice until the end, even I have been given some information which I didn't utilize, the probability remain the same, *i.e.* $1/3$. One should be careful it seems that after the revealing of one of the door by the host, only 2 doors left with a car behind so that probability of winning should be $1/2$, however, the restriction for the host to reveal a goat is a bias but not random selection, that's why $1/2$ in question (c) but not (a).

2. On the definition of countable sums

Recall that for a countable set Ω and a function $f : \Omega \rightarrow [0, \infty)$, we have defined the meaning of

$$\sum_{\omega \in \Omega} f(\omega) = \sup \left\{ \sum_{\omega \in F} f(\omega) : F \subseteq \Omega, F \text{ finite} \right\}$$

(with $\sum_{\omega \in \emptyset} f(\omega) = 0$) in Definition 3.1. It is clear that this definition is consistent with the definition of finite sums when Ω is finite. Show that the other two claims in Remark 3.2 hold:

- (a) Suppose in addition that Ω is infinite. Then for an enumeration of $\Omega = \{\omega_1, \omega_2, \dots\}$ (see Remark 3.2(ii)),

$$\sum_{\omega \in \Omega} f(\omega) = \sum_{j=1}^{\infty} f(\omega_j) = \lim_{N \rightarrow \infty} \sum_{j=1}^N f(\omega_j).$$

Conclude that the limit on the right-hand side does not depend on the choice of the enumeration.

- (b) If $(A_j)_{j \in \mathbb{N}}$ are pairwise disjoint with $\Omega = \bigcup_{j \in \mathbb{N}} A_j$, then

$$\sum_{\omega \in \Omega} f(\omega) = \sum_{j=1}^{\infty} \sum_{\omega \in A_j} f(\omega).$$

- (a) Note that for $F = \{\omega_1, \dots, \omega_n\}$ (without loss of generality) finite subset of Ω ,

$$\sum_{\omega \in F} f(\omega) \leq \sum_{j=1}^n f(\omega_j) \leq \sum_{j=1}^{\infty} f(\omega_j).$$

Taking the supremum over all the finite subset, we obtain

$$\sum_{\omega \in \Omega} f(\omega) \leq \sum_{j=1}^{\infty} f(\omega_j).$$

Hence one side of the equality is shown and we can assume $\sum_{\omega \in \Omega} f(\omega) < \infty$ because otherwise the right hand side will also be infinite by the above inequality. It suffices to show

$$\sum_{\omega \in \Omega} f(\omega) \geq \sum_{j=1}^{\infty} f(\omega_j)$$

This is clear by the definition of supremum, $\sum_{\omega \in \Omega} f(\omega) \geq \sum_{j=1}^N f(\omega_j)$ for all natural number N . Letting $N \rightarrow \infty$ shows the desired inequality. Combining two sides, we conclude the equality. Since the left-hand side defined by supremum is a fixed number, we conclude that the limit on the right-hand side does not depend on the choice of the enumeration.

- (b) Note that for F finite subset of Ω ,

$$\sum_{\omega \in F} f(\omega) = \sum_{j=1}^{\infty} \sum_{\omega \in A_j \cap F} f(\omega) \leq \sum_{j=1}^{\infty} \sum_{\omega \in A_j} f(\omega).$$

The first equality holds because the sums in the second term are in fact both finite sums. Take supremum over F , we obtain

$$\sum_{\omega \in \Omega} f(\omega) \leq \sum_{j=1}^{\infty} \sum_{\omega \in A_j} f(\omega).$$

Hence one side of the equality is shown and we can assume we are in the case $\sum_{\omega \in \Omega} f(\omega) < \infty$. It suffices to show that for any $\epsilon > 0$,

$$\sum_{\omega \in \Omega} f(\omega) \geq \sum_{j=1}^{\infty} \sum_{\omega \in A_j} f(\omega) - \epsilon.$$

Fix an $\epsilon > 0$. For every $j \in \mathbb{N}$, by the definition of summation over countable sets, there exists F_j finite such that

$$(*) \quad \sum_{\omega \in A_j} f(\omega) \leq \sum_{\omega \in F_j} f(\omega) + \frac{\epsilon}{2^j}.$$

We can sum over j up to a natural number N and see that

$$\sum_{j=1}^N \sum_{\omega \in A_j} f(\omega) \stackrel{(*)}{\leq} \sum_{j=1}^N \sum_{\omega \in F_j} f(\omega) + \sum_{j=1}^N \frac{\epsilon}{2^j} \leq \sum_{\omega \in \Omega} f(\omega) + \epsilon.$$

Finally, let $N \rightarrow \infty$ to conclude.

3. (*) Discrete distributions

[4 Points]

For (a) and (b), first determine which probability distribution you are using.

- (a) We roll a fair die until the first time we observe the number 6. What is the probability that we need more than 9 attempts?
- (b) In the production of a certain kind of computer chip defects occur independently with probability 0.01. How likely is it that there are at least 2 defects in a batch of 100 computer chips? Give the exact probability and compare it to the approximation with a Poisson distribution.

Show that the following approximation of probability mass functions holds:

- (c) The probability mass function of a hypergeometric distribution $H(N, M, n)$ with parameters $N, M, n \in \mathbb{N}$ converges to the probability mass function of a binomial distribution $\text{Bin}(n, p)$ as $N, M \rightarrow \infty$ if $\frac{M}{N} \rightarrow p$.

Consider the following thinning of a Poisson distribution:

- (d) Assume that the number of penalty kicks during a football match is Poisson distributed with some parameter $\lambda > 0$. Also assume that every penalty kick results in a goal with the same probability $p \in (0, 1)$, independently of all others. Determine the distribution of the number of goals obtained via a penalty kick.

Hint: For $k \in \mathbb{N}_0$, calculate the probability that there are exactly k goals obtained via a penalty kick.

- (a) The number of rolls of the die is geometrically distributed with success parameter $p = \frac{1}{6}$ ($\text{Geo}(\frac{1}{6})$) so

$$p(k) = p \cdot (1-p)^{k-1}, \quad k \in \mathbb{N}.$$

Here k corresponds to the number of rolls *including* the one when we obtain the number 6. Therefore,

$$\begin{aligned} \mathbf{P}[\text{"more than 9 rolls"}] &= \mathbf{P}[\{10, 11, 12, \dots\}] = \sum_{k=10}^{\infty} p(k) = p \sum_{k=9}^{\infty} (1-p)^k \\ &= \frac{1}{6} \cdot \left(\frac{5}{6}\right)^9 \cdot \sum_{k=0}^{\infty} \left(\frac{5}{6}\right)^k \\ &= \frac{1}{6} \cdot \left(\frac{5}{6}\right)^9 \cdot \frac{1}{1 - \frac{5}{6}} \approx 0.1938. \end{aligned}$$

- (b) The number of defects follows a binomial distribution with $n = 100$ and $p = 0.01$ ($\text{Bin}(100, 0.01)$), so

$$p(k) = \binom{100}{k} \cdot 0.01^k \cdot 0.99^{100-k}.$$

Therefore, the probability to find at least 2 chips with defects is

$$\begin{aligned} \mathbf{P}[\text{"at least 2 defects"}] &= \mathbf{P}[\{2, 3, 4, \dots, 100\}] = 1 - \mathbf{P}[\{0, 1\}] = 1 - p(0) - p(1) \\ &= 1 - 0.99^{100} - 100 \cdot 0.01 \cdot 0.99^{99} \\ &\approx 1 - 0.366 - 0.370 = 0.264. \end{aligned}$$

Note that by Proposition 3.6 in the notes, we can approximate the probability mass function of $\text{Bin}(n, p)$ for large n and small p by the probability mass function of the Poisson distribution with parameter $\lambda = n \cdot p$. Here, we approximate the probability mass function p by $\hat{p}(k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!}$, where $\lambda = 100 \cdot 0.01 = 1$. We obtain (with $\hat{\mathbf{P}} = \text{Poi}(1)$)

$$\begin{aligned} \hat{\mathbf{P}}[\text{"at least 2 defects"}] &= \hat{\mathbf{P}}[\{2, 3, 4, \dots\}] = 1 - \hat{\mathbf{P}}[\{0, 1\}] = 1 - \hat{p}(0) - \hat{p}(1) \\ &= 1 - e^{-1} - \frac{1^1}{1!} e^{-1} \\ &= 1 - \frac{2}{e} \approx 0.264. \end{aligned}$$

So the approximation is accurate to the first three digits.

- (c) The probability mass function of a hypergeometric distribution $H(N, M, n)$ is

$$\begin{aligned} p(k) &= \frac{\binom{M}{k} \cdot \binom{N-M}{n-k}}{\binom{N}{n}} = \frac{\frac{M!}{k!(M-k)!} \cdot \frac{(N-M)!}{(n-k)!(N-M-n+k)!}}{\frac{N!}{n!(N-n)!}} \\ &= \frac{n!}{k!(n-k)!} \cdot \frac{\frac{M!}{(M-k)!} \cdot \frac{(N-M)!}{(N-M-n+k)!}}{\frac{N!}{(N-k)!}} \\ &= \binom{n}{k} \frac{\prod_{i=0}^{k-1} (M-i) \cdot \prod_{i=0}^{n-k-1} (N-M-i)}{\prod_{i=0}^{n-1} (N-i)} \frac{\frac{1}{N^n}}{\frac{1}{N^n}} \\ &= \binom{n}{k} \cdot \frac{\prod_{i=0}^{k-1} (\frac{M}{N} - \frac{i}{N}) \prod_{i=0}^{n-k-1} (1 - \frac{M}{N} - \frac{i}{N})}{\prod_{i=0}^{n-1} (1 - \frac{i}{N})}, \end{aligned}$$

All products are finite and stay so for fixed k . We can therefore take the limit $M, N \rightarrow \infty$ inside the products. Note $M/N \rightarrow p$ and $i/N \rightarrow 0$ for fixed index i . So:

$$\begin{aligned} p(k) &\rightarrow \binom{n}{k} \cdot \frac{\prod_{i=0}^{k-1} (p - 0) \cdot \prod_{i=0}^{n-k-1} (1 - p - 0)}{\prod_{i=0}^{n-1} (1 - 0)} \\ &= \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k}, \quad \text{as } N, M \rightarrow \infty. \end{aligned}$$

This is indeed the probability mass function of $\text{Bin}(n, p)$.

- (d) We define the events

$$\begin{aligned} S_n &= \text{"}n\text{ penalty kicks occur during the match"} \quad (n \in \mathbb{N}_0) \\ T_k &= \text{"}k\text{ goals through penalty kicks occur during the match"} \quad (k \in \mathbb{N}_0). \end{aligned}$$

We consider a probability space $(\Omega, \mathcal{F}, \mathbf{P})$ on which these events can be defined. By assumption, we have that for $n \in \mathbb{N}$

$$\mathbf{P}[S_n] = \frac{\lambda^n}{n!} \cdot e^{-\lambda},$$

using that the number of penalty kicks follows a $Poi(\lambda)$ -distribution for $\lambda > 0$. Now if the number n of penalty kicks is known, the number of goals through a penalty kicks follows a binomial distribution with parameters n and p (since we assume that any penalty kick results in a goal with probability p , independently of the others). Moreover, the number of goals by penalty kick can only be as large as the number of total penalty kicks, so

$$\mathbf{P}[T_k|S_n] = \begin{cases} \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}, & k \leq n, \\ 0, & k > n. \end{cases}$$

Since $\bigcup_{n=0}^{\infty} S_n = \Omega$ and this union is disjoint, we can use the law of total probability (see Remark 2.8):

$$\begin{aligned} \mathbf{P}[T_k] &= \sum_{n=0}^{\infty} \mathbf{P}[T_k|S_n] \cdot \mathbf{P}[S_n] = \sum_{n=k}^{\infty} \binom{n}{k} \cdot p^k (1-p)^{n-k} \cdot \frac{\lambda^n}{n!} \cdot e^{-\lambda} \\ &= \frac{\lambda^k p^k e^{-\lambda}}{k!} \sum_{n=k}^{\infty} \underbrace{\frac{(\lambda(1-p))^{n-k}}{(n-k)!}}_{=e^{\lambda(1-p)}} = \frac{(\lambda p)^k}{k!} e^{-\lambda p}, \quad k \in \mathbb{N}_0. \end{aligned}$$

So the number of goals through penalty kicks again follows a Poisson distribution, with parameter λp .

4. n -step simple random walk and the ballot theorem

Let $n \in \mathbb{N}$. We define the set

$$\Omega_n = \{(\omega_j)_{j=0}^n \in \mathbb{Z}^n : \omega_0 = 0, |\omega_j - \omega_{j-1}| = 1, \text{ for all } j = 1, \dots, n\}.$$

We call the discrete uniform distribution on $(\Omega_n, \mathcal{P}(\Omega_n))$ the (law of the) n -step simple random walk starting from 0, and abbreviate it by P_0 .

- (a) Calculate $|\Omega_n|$. Explain in words why an outcome $\omega \in \Omega_n$ can be viewed as a trajectory of a simple random walk.
- (b) Consider for $k \in \mathbb{Z}$ the event

$$A_k = \{\omega \in \Omega_n : \omega_n = k\}.$$

Calculate $P_0[A_k]$. What is the interpretation of the event A_k ?

We consider now the following application: Suppose when voting for a certain motion A , $n \in \mathbb{N}$ ballots are cast. After all ballots are cast, they are counted one by one, and the result “for A ” and “against A ” is noted. We assume that it is known that A passes with $k \in \{1, \dots, n\}$ more “for A ” votes than “against A ” votes. We are interested in the probability that during the entire counting process, there are always (except at the beginning, when no ballots have been counted) more “for A ” votes than “against A ” votes.

- (c) Argue that this corresponds to calculating

$$P_0[\{\omega \in \Omega_n : \omega_j > 0, 1 \leq j \leq n\} | A_k],$$

and calculate the previous expression.

Hint: You need to find the number of $(n-1)$ -step paths from 1 to k that stay strictly positive. To do this, subtract from the total number of all $(n-1)$ -step paths from 1 to k the number of paths that attain a value ≤ 0 .

- (a) We note that Ω_n can be mapped bijectively to $\tilde{\Omega}_n = \{-1, 1\}^n$ via the mapping

$$\varphi_n : \Omega_n \rightarrow \tilde{\Omega}_n, \quad (\omega_0, \dots, \omega_n) \mapsto (\omega_1, \omega_2 - \omega_1, \dots, \omega_n - \omega_{n-1})$$

with inverse

$$\tilde{\varphi}_n : \tilde{\Omega}_n \rightarrow \Omega_n, \quad (\tilde{\omega}_1, \dots, \tilde{\omega}_n) \mapsto (0, \tilde{\omega}_1, \tilde{\omega}_1 + \tilde{\omega}_2, \dots, \tilde{\omega}_1 + \dots + \tilde{\omega}_n)$$

(the interpretation here is that $\tilde{\Omega}_n$ keeps track of the increments and Ω_n of the paths). Therefore, $|\Omega_n| = |\tilde{\Omega}_n| = 2^n$. The outcomes $\omega \in \Omega_n$ consist of paths that make a step upward or downward at every (discrete) time point, starting from 0.

(b) We distinguish in cases where $n \in 2\mathbb{N}$ (even) and $n \in 2\mathbb{N}_0 + 1$ (odd). First let $n \in 2\mathbb{N}$, then

$$\begin{aligned} P_0[A_k] &= \frac{1}{2^n} |\{\omega \in \Omega_n : \omega_n = k\}| \\ &= \begin{cases} 0, & k \in 2\mathbb{Z} + 1, \\ \frac{1}{2^n} \binom{n}{(n+k)/2}, & k \in 2\mathbb{Z} \end{cases} \end{aligned}$$

since in order for the string $\omega = (\omega_0, \omega_1, \dots, \omega_n)$ to end in k , there must be k more steps up than down, i.e. $(n+k)/2$ steps are “up” steps ($\omega_j - \omega_{j-1} = +1$) and $(n-k)/2$ steps are “down” steps ($\omega_j - \omega_{j-1} = -1$).

Similarly, if $n \in 2\mathbb{N}_0 + 1$ is odd, we have

$$P_0[A_k] = \begin{cases} 0, & k \in 2\mathbb{Z}, \\ \frac{1}{2^n} \binom{n}{(n+k)/2}, & k \in 2\mathbb{Z} + 1 \end{cases}$$

(c) We can view a “vote for A ” at time $j \in \mathbb{N}$ as a step up ($\omega_j - \omega_{j-1} = +1$) and a “vote against A ” ($\omega_j - \omega_{j-1} = -1$) at time $j \in \mathbb{N}$ as a step down. Therefore, the difference in votes for and against A corresponds to the trajectory $\omega \in \Omega_n$. Since it is known that in the end A passes with k more “for” votes than “against” votes, this corresponds to all paths ω with $\omega_n = k$, which is exactly the event A_k .

On the other hand, the fact that A is always in the lead corresponds to the difference always being (strictly) positive, which is the event $\{\omega \in \Omega_n : \omega_j > 0, 1 \leq j \leq n\}$.

We thus calculate

$$\begin{aligned} P_0[\{\omega \in \Omega_n : \omega_j > 0, 1 \leq j \leq n\} | A_k] &= \frac{\frac{1}{2^n} |\{\omega \in \Omega_n : \omega_j > 0, 1 \leq j \leq n\} \cap A_k|}{\frac{1}{2^n} \binom{n}{(n+k)/2}} \\ &= \frac{|\{\omega \in \Omega_n : \omega_j > 0, 1 \leq j \leq n\} \cap A_k|}{\binom{n}{(n+k)/2}}, \end{aligned}$$

if either n and k are both even or both odd (note that this is implicitly required: A can only pass with k more “for” than “against” votes, if $n - k$ is even).

We now calculate the cardinality in the numerator: Clearly, for ω to be in the set $\{\omega \in \Omega_n : \omega_j, 1 \leq j \leq n\}$, we must have $\omega_1 = +1$. We therefore have (as suggested in the hint):

$$|\{\omega \in \Omega_n : \omega_j > 0, 1 \leq j \leq n\} \cap A_k| = r - s,$$

where

$$\begin{aligned} r &= \text{the number of paths from } (1, 1) \text{ to } (n, k) = \\ &= \text{the number of paths from } (0, 0) \text{ to } (n-1, k-1), \\ s &= \text{the number of paths from } (1, 1) \text{ to } (n, k) \text{ that ever attain value } 0 \\ &= \text{the number of paths from } (0, 0) \text{ to } (n-1, k-1) \text{ that ever attain value } -1 \end{aligned}$$

For r one argues in exactly the same way as in (b) that

$$r = \binom{n-1}{\frac{n+k-2}{2}}.$$

To find the number of $n-1$ step paths starting in $(0, 0)$ that attain the value -1 , we can define for such a path $\omega \in \Omega_{n-1}$

$$j_* = \inf\{1 \leq j \leq n : \omega_j = -1\}.$$

By reflecting a path ω from $(0, 0)$ to $(n-1, k-1)$ that visits -1 , i.e.

$$\omega = (0, \dots, \omega_{j_*-1}, \underbrace{\omega_{j_*}}_{=-1}, \omega_{j_*+1}, \dots, \underbrace{\omega_{n-1}}_{k-1})$$

after j_* at the line $y = -1$, one obtains another path

$$\omega_{refl} = (0, \dots, \omega_{j_*-1}, \underbrace{\omega_{j_*}}_{=-1}, -2 - \omega_{j_*+1}, \dots, \underbrace{-2 - \omega_n}_{=-k-1})$$

that ends in $-k-1$. It is easy to see that the path from $(0, 0)$ to $(n-1, k-1)$ that visit -1 are in bijection to *all* paths from $(0, 0)$ to $(n-1, -k-1)$. (This is also called the *reflection principle*).

Using this we can finally calculate

$$s = \text{the number of paths from } (0, 0) \text{ to } (n-1, -k-1) = \binom{n-1}{\frac{n-k-2}{2}}.$$

Putting things together gives us

$$\begin{aligned} r - s &= \binom{n-1}{\frac{n+k-2}{2}} - \binom{n-1}{\frac{n-k-2}{2}} = \frac{(n-1)!}{\left(\frac{n+k}{2}-1\right)! \left(\frac{n-k}{2}\right)!} - \frac{(n-1)!}{\left(\frac{n-k}{2}-1\right)! \left(\frac{n+k}{2}\right)!} \\ &= \frac{n}{n} \frac{(n-1)! \frac{n+k}{2}}{\left(\frac{n+k}{2}\right)! \left(\frac{n-k}{2}\right)!} - \frac{n}{n} \frac{(n-1)! \frac{n-k}{2}}{\left(\frac{n-k}{2}\right)! \left(\frac{n+k}{2}\right)!} = \frac{1}{n} \binom{n}{\frac{n+k}{2}} \frac{n+k}{2} - \frac{1}{n} \binom{n}{\frac{n+k}{2}} \frac{n-k}{2} \\ &= \binom{n}{\frac{n+k}{2}} \frac{k}{n}, \end{aligned}$$

and finally:

$$P_0[\{\omega \in \Omega_n : \omega_j > 0, 1 \leq j \leq n\} | A_k] = \frac{r-s}{\binom{n}{(n+k)/2}} = \frac{k}{n}.$$

This last part is called the *ballot theorem*.